Onshore wind energy land impacts: Implications for Atlantic Canada using the Atlantic Canada Energy System Model (ACES)

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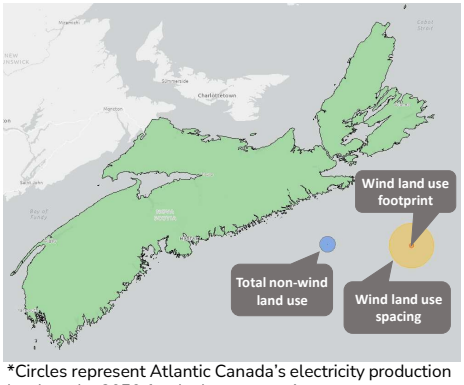
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Abstract

- This work examines the interconnections of electricity generation costs, emissions and land footprints.
- Our objective is to find an optimized energy mix for transition to a sustainable energy system while meeting the net zero emission targets.
- We studied Atlantic Canada with the Atlantic Canada Energy System model (ACES).
- We compare a base scenario with a limited emissions (CO2e) case and a land limit (+emissions limit) scenario as seen in the table.

Scenario Assumptions	Base	Emission Limit	Land Limit
Carbon Tax	✓	✓	✓
Coal Phaseout	✓	✓	✓
Net Zero by 2050	✗	✓	✓
Year:	2030	2035	2040	2045	2050
CO2e(Mt):	2	1.5	1	0.5	0
Total Land Use	2770 km2	2950 km2	2830 km2
Total Wind Land Use	2470 km2	2600 km2	2470 km2
Wind Land Use(New)	2030 km2	2160 km2	2030 km2
Land Limit	✗	✗	✓

*land limit is only applied on new onshore wind build-out and is equal to this amount for the base scenario.



*Circles represent Atlantic Canada's electricity production land use by 2050 for the base scenario.

Introduction

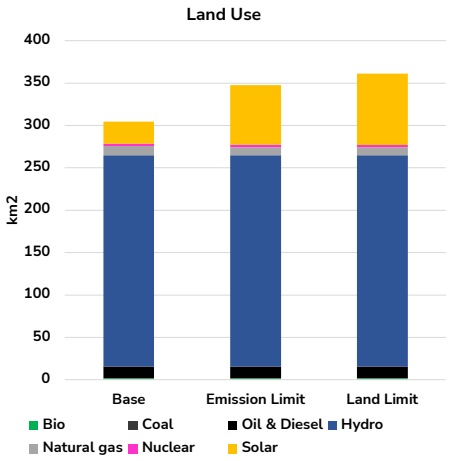
- Renewables apply a major land footprint to the energy system due to their high land intensity and extensive development [1].
- Onshore wind has the highest land intensity among other energy production technologies if we consider the spacing between turbines [2].
- The project objective is to showcase the energy system under land constraints and investigate net zero pathways.

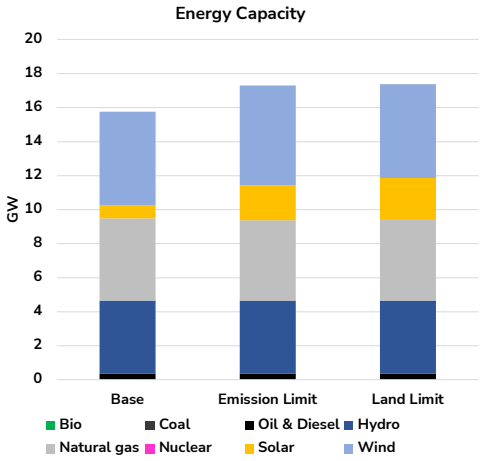
Methods

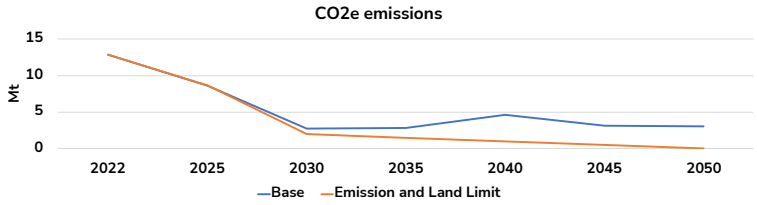
- We worked on Atlantic Canada as our case study to investigate land use implications. For this purpose, we used the ACES (Atlantic Canada Energy System) model [3].
- ACES is a capacity expansion model based on cost optimization. It is based on the open-source TEMOA (Tools for Energy Modeling Optimization Analysis) model [4].
- We first applied a post-processing approach to the electricity system of our case study to track the land use.
- We then ran a net zero by 2050 scenario. Finally, we limited the onshore wind land use of the emission limit scenario to that of the base scenario and ran a land limit scenario.

Results

- A decrease in available land for onshore wind results in an increase in solar power. This choice is due to the lower land intensity of utility-scale solar compared to onshore wind (34.4 vs 368.9 km2)
- Overall costs of the system increase by 3 percent for both limitation scenarios, however, the difference between the two scenarios is not significant.
- While emissions reached net zero following the constraints, it was forced by applying negative emissions through CO2 capture technologies. These technologies had a slightly higher contribution to the land limit scenario.
- Both emission and land limits cause changes in the inter-provincial transmissions and minimize new transmission line build-out.







Conclusion

- ✓ Policies and decisions should be based on multiple factors, looking beyond costs and emissions. Therefore, it is important to look at land use implications of different scenarios, considering various constraints, such as zero-emission cases.
- ✓ Land intensity of different energy production technologies plays a pivotal role in energy system design, however, focusing on differences between metrics such as footprint and spacing is important while interpreting model results.
- ✓ Future research plans should include a more comprehensive suitability analysis on wind farm siting. Factors such as local wind potentials and current land cover greatly impact the generation potential and land occupation consequences. This indicates that more local studies are required to investigate imports and transmissions between regions and consider more impacting factors.
- ✓ Future research should also conduct scenarios with more aggressive land constraints to observe their impacts.

References

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[2] K. Palmer-Wilson et al., "Impact of land requirements on electricity system decarbonisation pathways," Energy Policy, vol. 129, pp. 193–205, Jun. 2019.

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